Lab Report

Router-on-a-Stick Style Multi-LAN Configuration with DHCP, DNS, and Web Server Deployment in Cisco Packet Tracer

Course: Computer Networking Lab

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1. Objectives

- To design and simulate a multi-LAN network using a single router (Cisco 2911) with two separate subnets in Cisco Packet Tracer.
- To configure two router interfaces (Gig0/0 and Gig0/1) with distinct IP addresses acting as default gateways for two different LANs.
- To configure the router as a DHCP server with two separate DHCP pools, one for each LAN, including excluded addresses, default gateway, and DNS server options.
- To deploy and statically configure a DNS Server and a Web Server on one LAN, and to verify domain name resolution and web access from client machines on both LANs.
- To verify successful DHCP address allocation and end-to-end connectivity across both LANs, including a remotely connected laptop.

2. Requirements / Tools Used

- Cisco Packet Tracer (network simulation software)
- 1 x Router (Cisco 2911)
- 2 x Switch (Cisco 2960-24TT)
- 6 x PC (PC-PT), 1 x Laptop (Laptop-PT)
- 1 x DNS Server (Server-PT), 1 x Web Server (Server-PT)
- Copper straight-through and cross-over cables as required

3. Network Topology

The topology consists of a single Cisco 2911 router connecting two separate LAN segments through its two Gigabit Ethernet interfaces. Interface Gig0/1 serves the newly added LAN (192.168.10.0/25), which hosts PC0, PC1, PC2, and a remotely connected Laptop1. Interface Gig0/0 serves the original LAN (172.17.0.0/27), which hosts PC3, PC4, PC5, a DNS Server, and a Web Server. Both switches (2960-24TT) interconnect the respective end devices to the router.

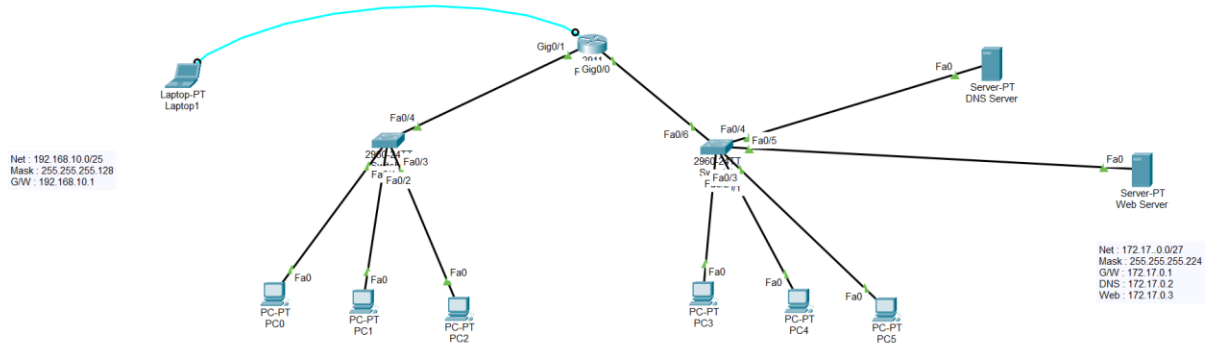


Figure 1: Overall LAN topology showing the router, two switches, six PCs, one remote laptop, DNS server, and web server.

3.1 IP Addressing Plan

Parameter	Value
Network	192.168.10.0/25 (Gig0/1 — new LAN)
Subnet Mask	255.255.255.128
Default Gateway	192.168.10.1
DNS Server (used)	172.17.0.2

Parameter	Value
Network	172.17.0.0/27 (Gig0/0 — original LAN)
Subnet Mask	255.255.255.224
Default Gateway	172.17.0.1
DNS Server	172.17.0.2
Web Server	172.17.0.3

4. Configuration Procedure

4.1 Configuring Router Interface Gig0/0 (Original LAN)

The Gig0/0 interface, connecting to the original LAN, was configured with the IP address 172.17.0.1/27 as verified from the router terminal accessed through the connected laptop.

```
.
interface GigabitEthernet0/0
 ip address 172.17.0.1 255.255.255.224
 duplex auto
 speed auto
```

Figure 2: Gig0/0 interface configuration (172.17.0.1 255.255.255.224).

```
interface GigabitEthernet0/0
```

```
ip address 172.17.0.1 255.255.255.224
duplex auto
speed auto
```

4.2 Configuring Router Interface Gig0/1 and DHCP for the New LAN

Interface Gig0/1 was enabled and assigned the IP address 192.168.10.1/25 to serve as the gateway for the newly added LAN. A DHCP pool named “Seu1” was then created for this subnet, with the router's own gateway address excluded from the DHCP scope and the DNS server address (172.17.0.2, located on the other LAN) assigned as the DNS option.

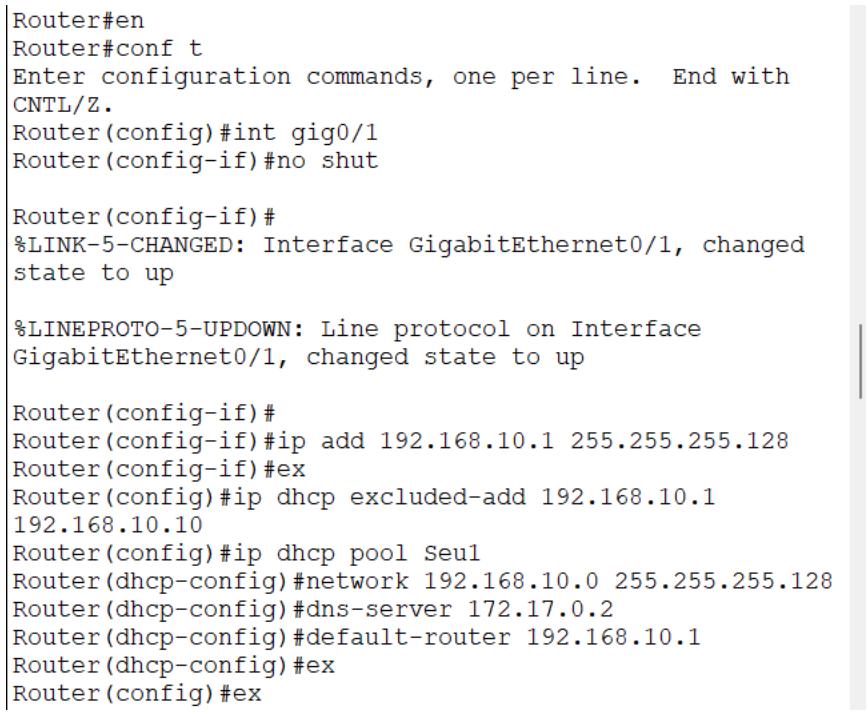


Figure 3: Router terminal output showing Gig0/1 activation, IP addressing, DHCP exclusion, and DHCP pool “Seu1” creation for the 192.168.10.0/25 network.

```
Router#en
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#int gig0/1
Router(config-if)#no shut

%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1,
changed state to up

Router(config-if)#ip add 192.168.10.1 255.255.255.128
Router(config-if)#ex
Router(config)#ip dhcp excluded-add 192.168.10.1 192.168.10.10
Router(config)#ip dhcp pool Seu1
Router(dhcp-config)#network 192.168.10.0 255.255.255.128
Router(dhcp-config)#dns-server 172.17.0.2
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#ex
```

4.3 DHCP Pool Configuration Summary (Original LAN)

A similar DHCP configuration was applied for the original LAN on the 172.17.0.0/27 network: the network range, default router, and DNS server address were defined within the DHCP pool, and the first three addresses (router and server addresses) were excluded from the DHCP scope.

```
:
ip dhcp excluded-address 172.17.0.1 172.17.0.3
!
ip dhcp pool Seu1
 network 172.17.0.0 255.255.255.224
 default-router 172.17.0.1
 dns-server 172.17.0.2
!
!
```

Figure 4: DHCP pool "Seu1" configuration for network 172.17.0.0/27 with excluded addresses, default router, and DNS server.

```
!
ip dhcp excluded-address 172.17.0.1 172.17.0.3
!
ip dhcp pool Seu1
 network 172.17.0.0 255.255.255.224
 default-router 172.17.0.1
 dns-server 172.17.0.2
!
```

4.4 Verifying Both DHCP Exclusions

The running configuration was checked to confirm that excluded addresses were correctly applied for both DHCP pools before the DHCP service began leasing addresses to clients.

```
Router#show run
Building configuration...

Current configuration : 926 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
!
!
!
ip dhcp excluded-address 172.17.0.1 172.17.0.3
ip dhcp excluded-address 192.168.10.1 192.168.10.10
!
ip dhcp pool Seu1
 network 192.168.10.0 255.255.255.128
 default-router 192.168.10.1
 dns-server 172.17.0.2
!
```

Figure 5: Router running-configuration excerpt confirming excluded addresses for both the 172.17.0.0/27 and 192.168.10.0/25 DHCP pools.

```
Router#show run
Building configuration...

Current configuration : 926 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname Router
!
ip dhcp excluded-address 172.17.0.1 172.17.0.3
ip dhcp excluded-address 192.168.10.1 192.168.10.10
!
ip dhcp pool Seu1
 network 192.168.10.0 255.255.255.128
 default-router 192.168.10.1
 dns-server 172.17.0.2
!
```

4.5 Final Interface Configuration

Both Gigabit Ethernet interfaces were verified together, confirming Gig0/0 as 172.17.0.1/27 and Gig0/1 as 192.168.10.1/25, both active with auto duplex and auto speed.

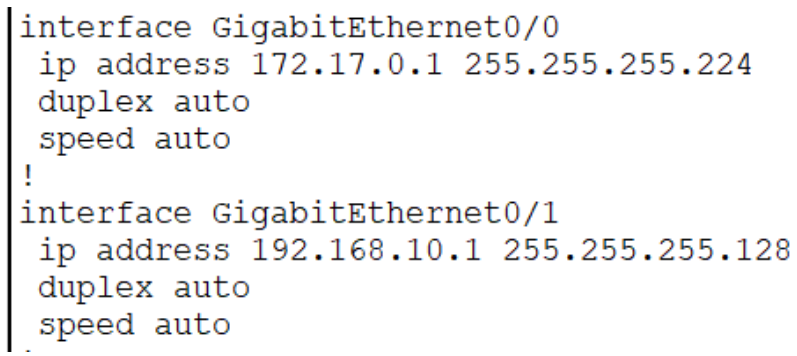


Figure 6: Combined interface configuration for Gig0/0 (172.17.0.1/27) and Gig0/1 (192.168.10.1/25).

```
interface GigabitEthernet0/0
 ip address 172.17.0.1 255.255.255.224
 duplex auto
 speed auto
!
interface GigabitEthernet0/1
 ip address 192.168.10.1 255.255.255.128
 duplex auto
 speed auto
```

5. IP Configuration of Servers

Since the DNS Server and Web Server must be reliably reachable at fixed addresses, both were assigned static IP configuration instead of DHCP, using addresses reserved (excluded) on the router.

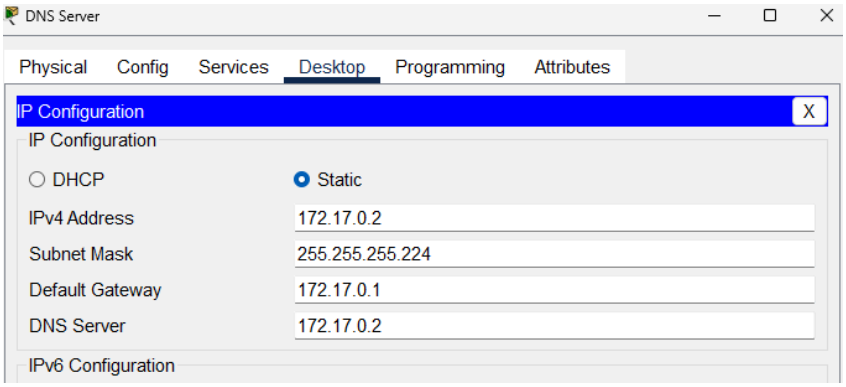


Figure 7: Static IP configuration of the DNS Server (172.17.0.2 / 255.255.255.224, gateway 172.17.0.1).

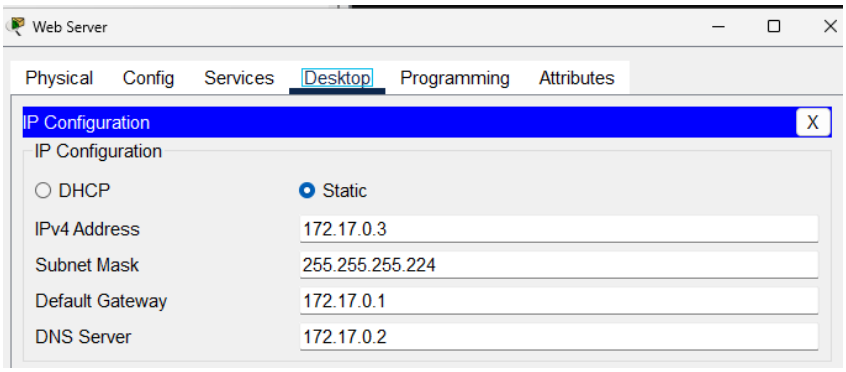


Figure 8: Static IP configuration of the Web Server (172.17.0.3 / 255.255.255.224, gateway 172.17.0.1, DNS 172.17.0.2).

6. Verification and Testing

6.1 Web Access Verification from the New LAN

From PC2 (192.168.10.0/25 subnet), the web browser was used to request http://www.sajjad.com. The page loaded successfully, confirming that DNS resolution via 172.17.0.2 and routing between the two LANs through the router were both functioning correctly.

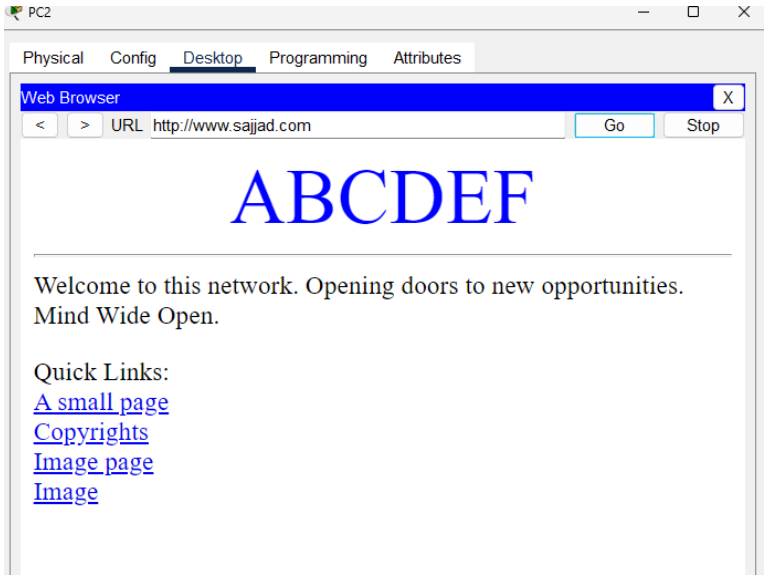


Figure 9: PC2 (new LAN) successfully accessing www.sajjad.com hosted on the Web Server.

6.2 DHCP Address Allocation Verification

DHCP was verified on client machines from both LANs. PC1, on the 192.168.10.0/25 network, successfully obtained an IP address (192.168.10.12) with the correct gateway (192.168.10.1) and DNS server (172.17.0.2). PC4, on the 172.17.0.0/27 network, successfully obtained an IP address (172.17.0.5) with the correct gateway (172.17.0.1) and DNS server (172.17.0.2).

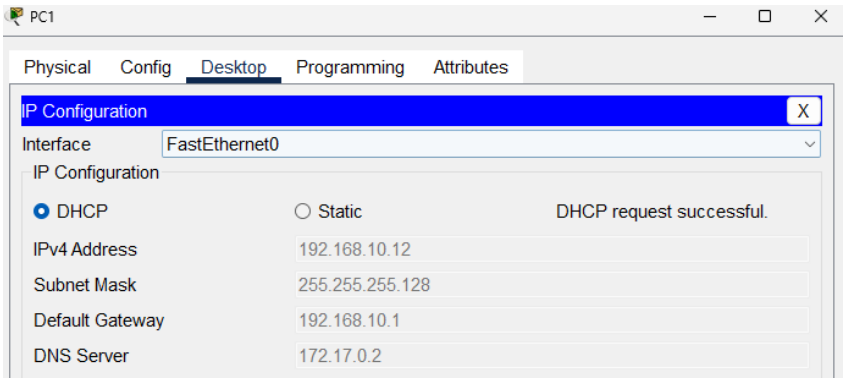


Figure 10: PC1 IP configuration showing successful DHCP lease (192.168.10.12) on the new LAN.

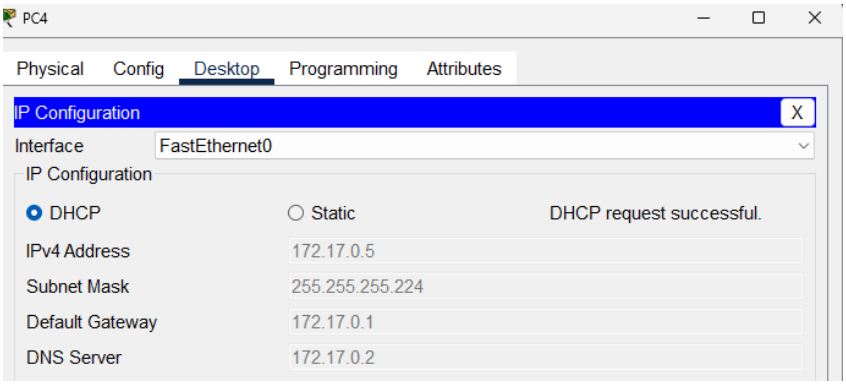


Figure 11: PC4 IP configuration showing successful DHCP lease (172.17.0.5) on the original LAN.

6.3 Final Web Access Verification

As a final check, PC4 on the original LAN also accessed http://www.sajjad.com successfully, confirming that the web service is reachable from clients on both subnets.

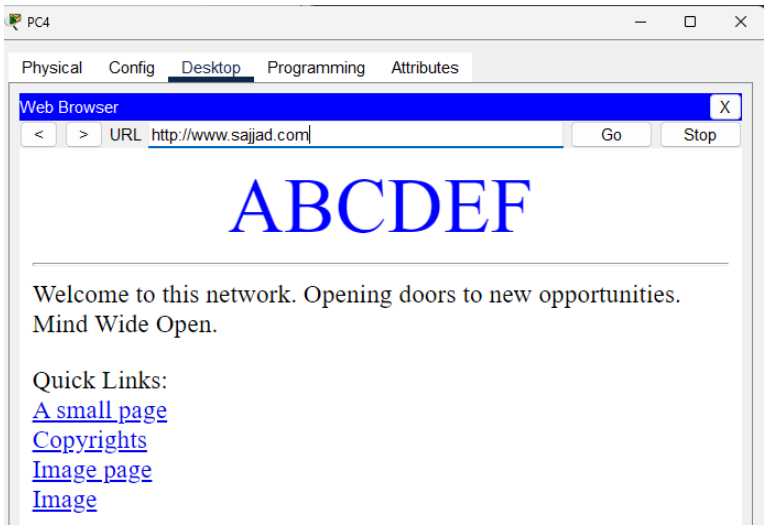


Figure 12: PC4 (original LAN) successfully accessing www.sajjad.com, confirming full network functionality.

7. Result

The multi-LAN network was successfully designed and configured in Cisco Packet Tracer. The router was configured with two Gigabit Ethernet interfaces acting as gateways for two separate subnets (192.168.10.0/25 and 172.17.0.0/27), and two independent DHCP pools were created to automatically assign IP addresses, default gateways, and DNS server information to client devices on each subnet. The DNS Server and Web Server were statically configured and successfully resolved and served the domain www.sajjad.com to clients on both LANs. DHCP verification on PC1 and PC4 confirmed correct automatic addressing, and browser-based access tests from PC2 and PC4 confirmed full inter-LAN connectivity and functioning DNS resolution. All configured objectives were achieved.

8. Conclusion

This lab provided practical experience in configuring a router to support multiple LANs on separate subnets, deploying router-based DHCP services with pool exclusions, and integrating DNS and Web servers into a routed network. The successful end-to-end verification across both LANs demonstrates a solid understanding of inter-VLAN style routing, DHCP scope management, and client-server name resolution in a simulated enterprise environment.